

Dr. Dipankar Bhattacharya

Marie Curie Fellow | Robotics, Control Systems & AI

Imperial College London | Dyson School of Engineering

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15+	10+	5+	8+
Publications	Research Projects	Countries	Years Experience

Research Domains

Medical Robotics

Soft robotics, rehabilitation systems, exoskeletons, and assistive devices for healthcare applications.

Industrial Automation

Fabric manipulation, garment production automation, robotic handling systems.

Cable-Driven Robotics

Parallel robots, vine robots, modeling and control of cable-driven systems.

AI & Machine Learning

Deep learning, imitation learning, reinforcement learning for robotic control.

Current Research

CASREx: Cable-driven Arm Soft Re-configurable Exoskeleton

Marie Skłodowska-Curie Fellowship Project

Imperial College London | Nov 2025 - Current

Developing innovative cable-driven soft exoskeleton systems for muscle impairment rehabilitation, combining advanced control systems with soft robotics principles.

Employment History

Marie Skłodowska-Curie Postdoctoral Fellow

Nov 2025 - Current

Dyson School of Engineering, Imperial College London

Supervisor: Prof. Thrishantha Nanayakkara

Project: CASREx - Cable-driven Arm Soft Re-configurable Exoskeleton for Muscle Impairment Rehabilitation

Senior Research Engineer - Control Systems

May 2025 - Oct 2025

Center for Transformative Garment Production (TransGP)

Supervisor: Prof. Kazuhiro Kosuge

Project: Physics-based Cloth Manipulation with policy generated from imitation learning

- Led multidisciplinary team developing high-performance models for fabric handling and folding
- Investigated feasibility and oversaw development of robotic systems for fabric manipulation

- Translated industrial requirements into research deliverables
- Mentored 3 PhD students

Postdoctoral Fellow

Jul 2024 - Apr 2025

Center for Transformative Garment Production (TransGP)

Supervisor: Prof. Kazuhiro Kosuge

Visiting Research Associate

Jul 2024 - Oct 2025

Department of Electrical and Electronic Engineering, Hong Kong University (HKU)

Postdoctoral Fellow*Mar 2021 - Jun 2024**Department of Mechanical and Automation, The Chinese University of Hong Kong***Supervisor: Prof. Darwin Lau****Projects:**

- Modeling and control of cable-driven robots with cable wrapping around obstacles and links
- Design of a cable-driven growing vine robot for safer false ceiling inspections

Responsibilities:

- Conducted independent and collaborative research
- Wrote research articles, proposals, and technical reports
- Mentored undergraduate and postgraduate students
- Delivered lectures for MAEG5090 Topics in Robotics

Visiting Postdoctoral Fellow*Oct 2021 - Jan 2022**Laboratoire des Sciences du Numérique de Nantes (LS2N), École Centrale de Nantes***Supervisors: Prof. Stephane Caro and Prof. Darwin Lau****Project:** Iterative-learning control of cable-driven parallel robot**Graduate Teaching Assistant***2017 - 2020**The University of Auckland***Courses:** MECHENG 706, MECHENG 306, ENGGEN 115, ENGGEN 131

- Supervised undergraduate students on design projects
- Delivered tutorials, workshops, and drop-in clinics
- Managed laboratory sessions and coordinated TA activities

Lecturer (Assistant Professor Grade-1)*2013 - 2016**School of Electrical, Electronics and Communication Engineering, Galgotia's University*

- Delivered lectures for undergraduate and postgraduate courses
- Developed teaching materials and laboratory sessions
- Supervised final year undergraduate projects

Education

Ph.D., Mechatronics Engineering*2016 - 2021**Department of Mechanical and Mechatronics Engineering, The University of Auckland***Thesis:** *Robotic Soft Esophagus Design, Modelling, and Control for Endoprosthetic Stent Testing and Food Swallow Investigation***M.Tech, Systems and Control***2011 - 2013**Department of Electrical Engineering, Indian Institute of Technology (IIT) Roorkee***Dissertation:** *Non-Linear Predictive Control Model using Relevance Vector Machine***GPA:** 8.4 / 10.0**B.Tech, Electronics and Communication Engineering***2004 - 2010**North Eastern Regional Institute of Science and Technology (NERIST)***Project:** *Design and Synthesis of a Sixth Order Tow Thomas Bi-quad filter***GPA:** 9.1 / 10.0

Technical Skills

Programming Languages

Python, MATLAB, C/C++, LabVIEW, Assembly

AI & ML Frameworks

PyTorch, TensorFlow, Gym, Blender

Robotics & Control

ROS, MATLAB/Simulink, Control Systems, Path Planning

CAD & Design

SOLIDWORKS, Inventor, Blender, Inkscape

Hardware

Raspberry Pi, Arduino, myRIO, 8051

Tools & Others

Git, L^AT_EX, Lightroom, Linux

Languages

English (Fluent), Hindi (Native)

Selected Publications

15+ peer-reviewed publications in international journals and conferences

Modeling and Control of Cable-Driven Robots with Cable Wrapping

D. Bhattacharya, D. Lau, et al.

IEEE Transactions on Robotics

Design of Cable-Driven Growing Vine Robot for Ceiling Inspection

D. Bhattacharya, D. Lau, et al.

IEEE/ASME International Conference on Advanced Intelligent Mechatronics

Robotic Soft Esophagus for Stent Testing Applications

D. Bhattacharya, P. Xu, et al.

Journal of Medical Devices

Iterative Learning Control for Cable-Driven Parallel Robots

D. Bhattacharya, S. Caro, D. Lau

Mechanism and Machine Theory

Physics-based Cloth Manipulation with Imitation Learning

D. Bhattacharya, K. Kosuge, et al.

IEEE International Conference on Robotics and Automation (ICRA)

For complete list of publications, please visit my [portfolio website](#)